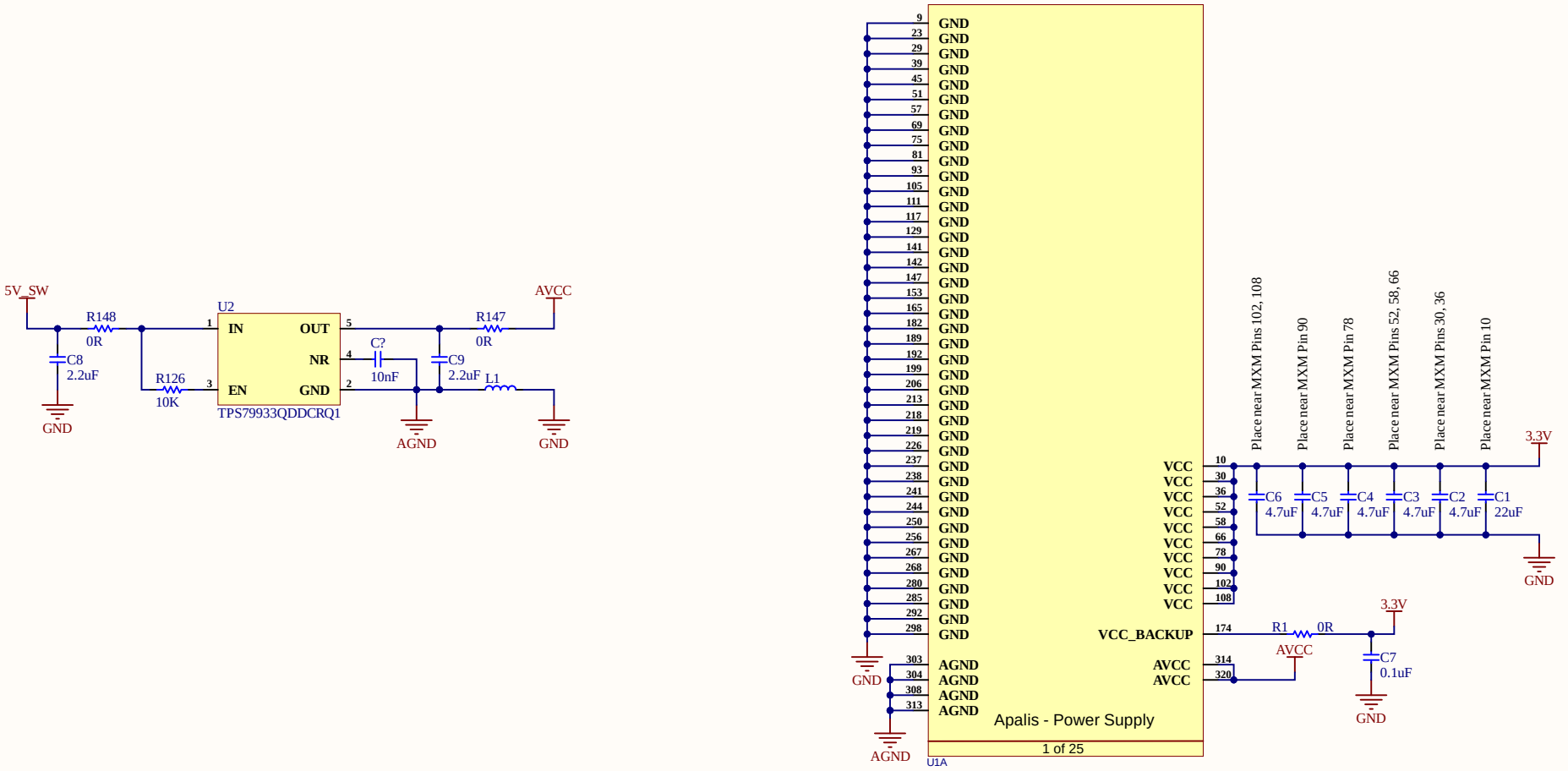
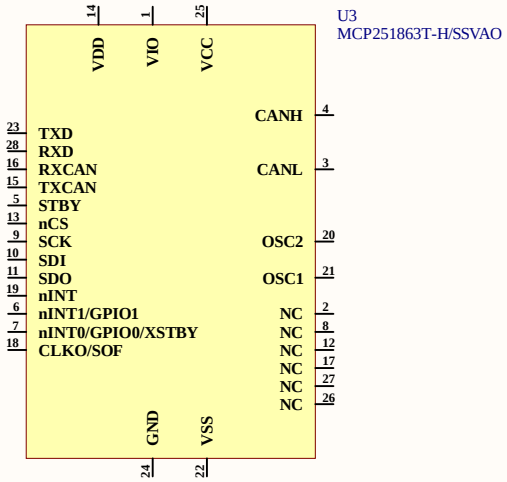
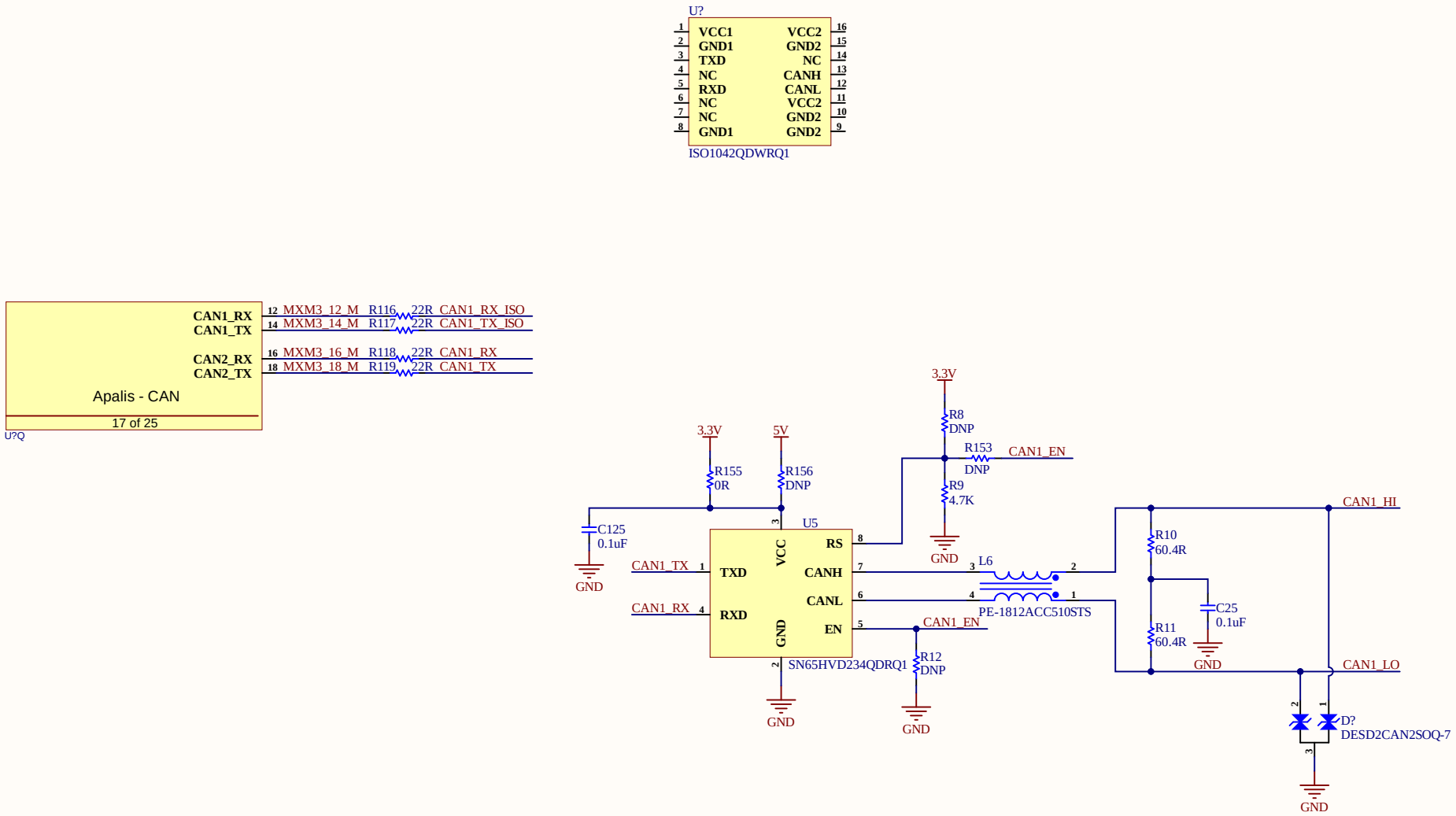
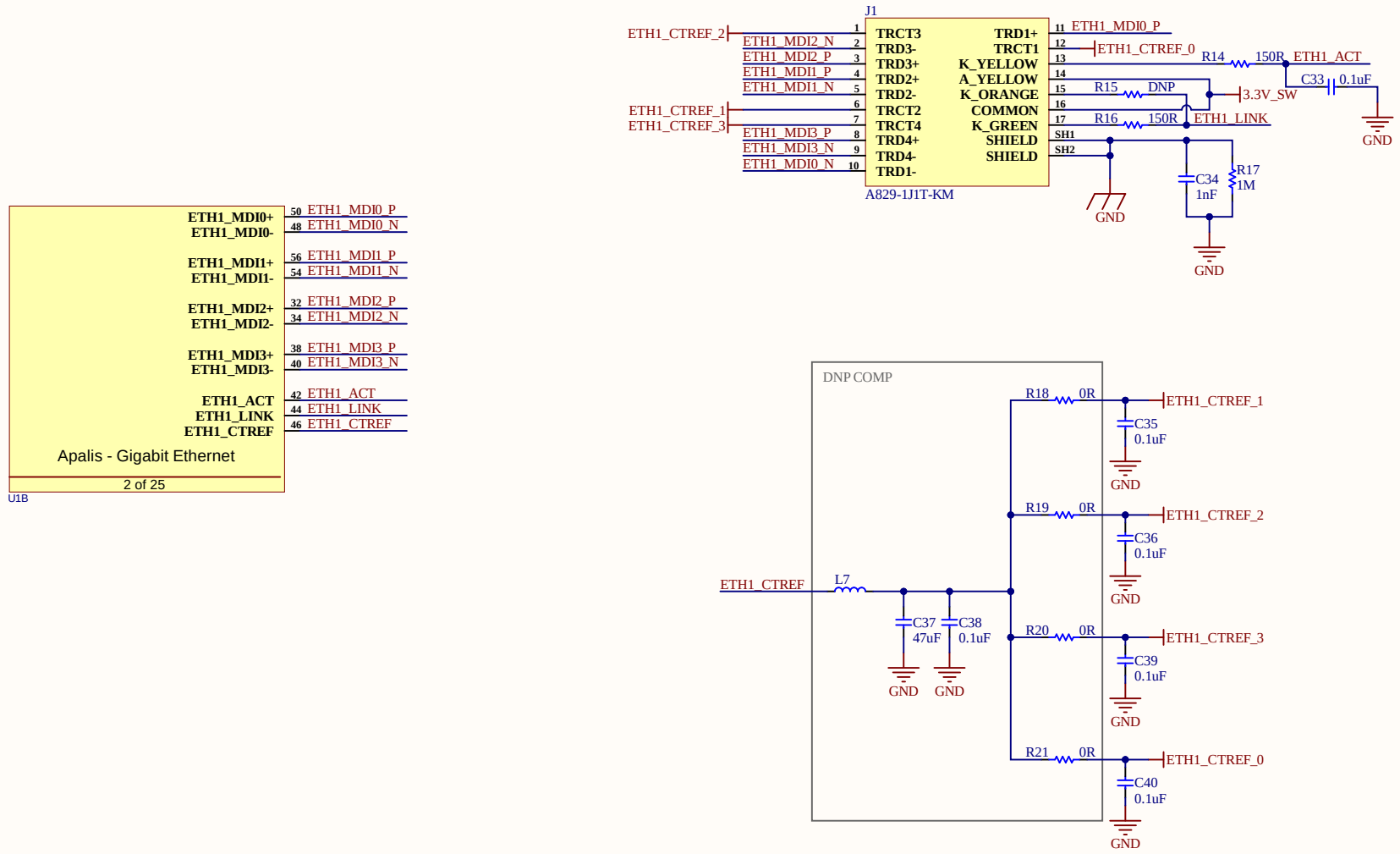
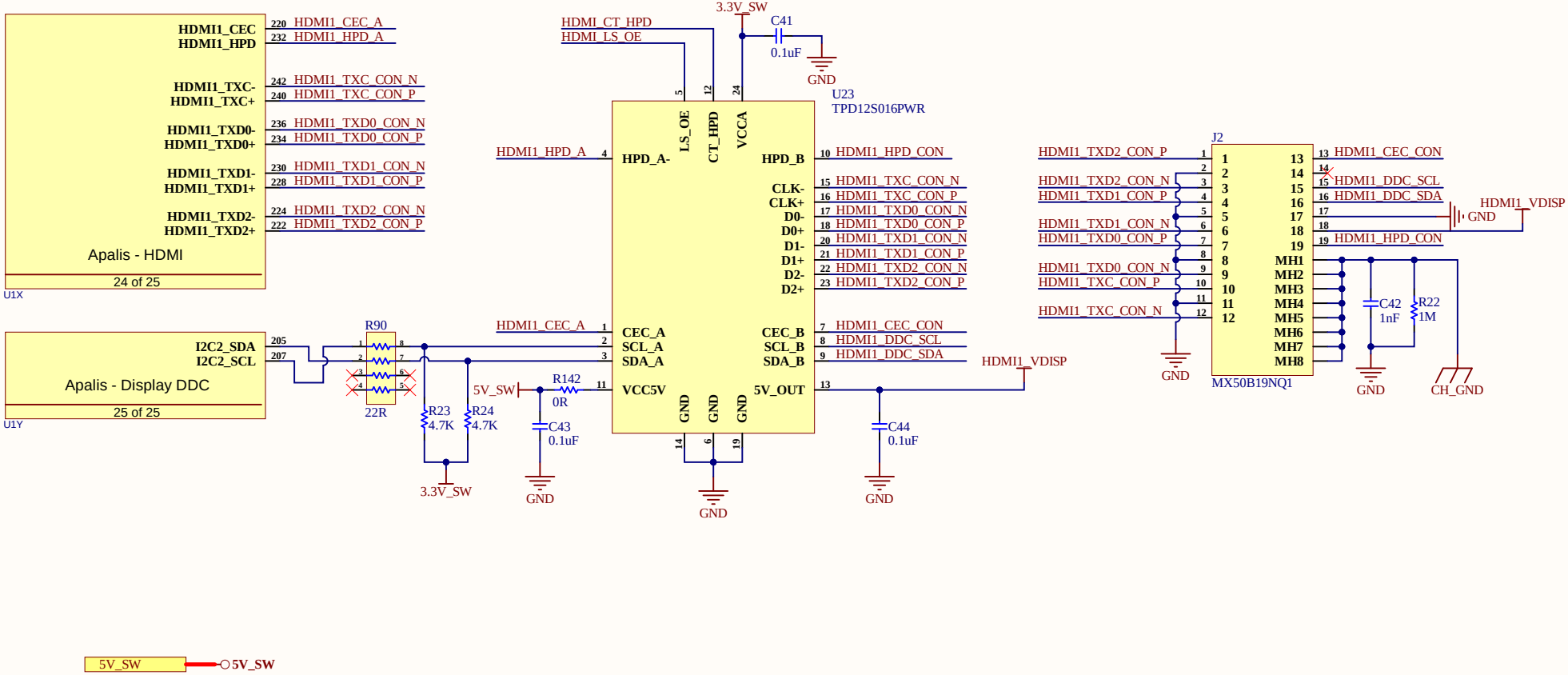
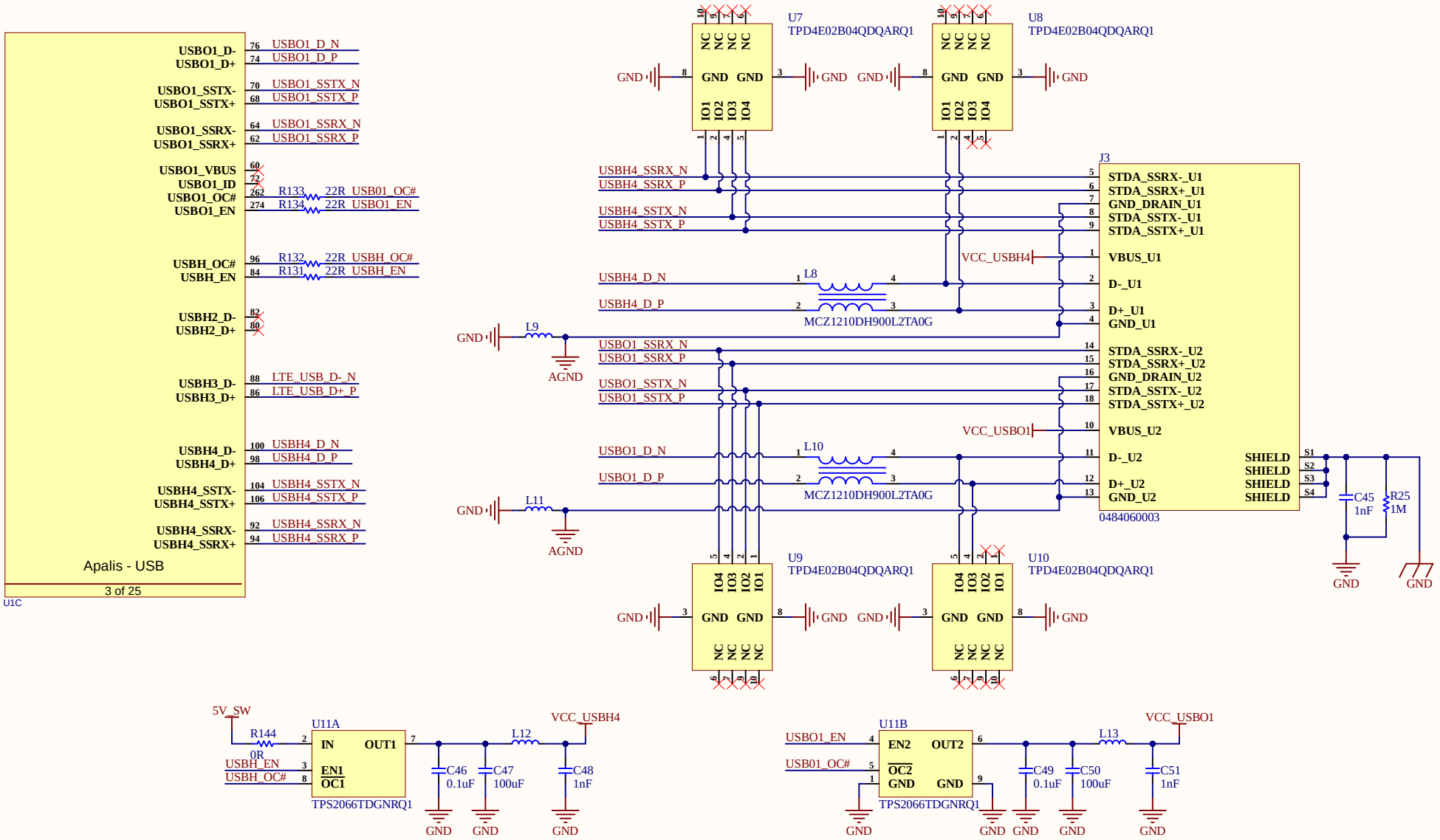

5V_SW
R148
0R
C8
2.2uF
GND
U2
IN
OUT
NR
EN
GND
TPS79933QDDCRQ1
R126
10K
C?
10nF
C9
2.2uF
AGND
R147
0R
L1
AVCC
GND
GND
GND

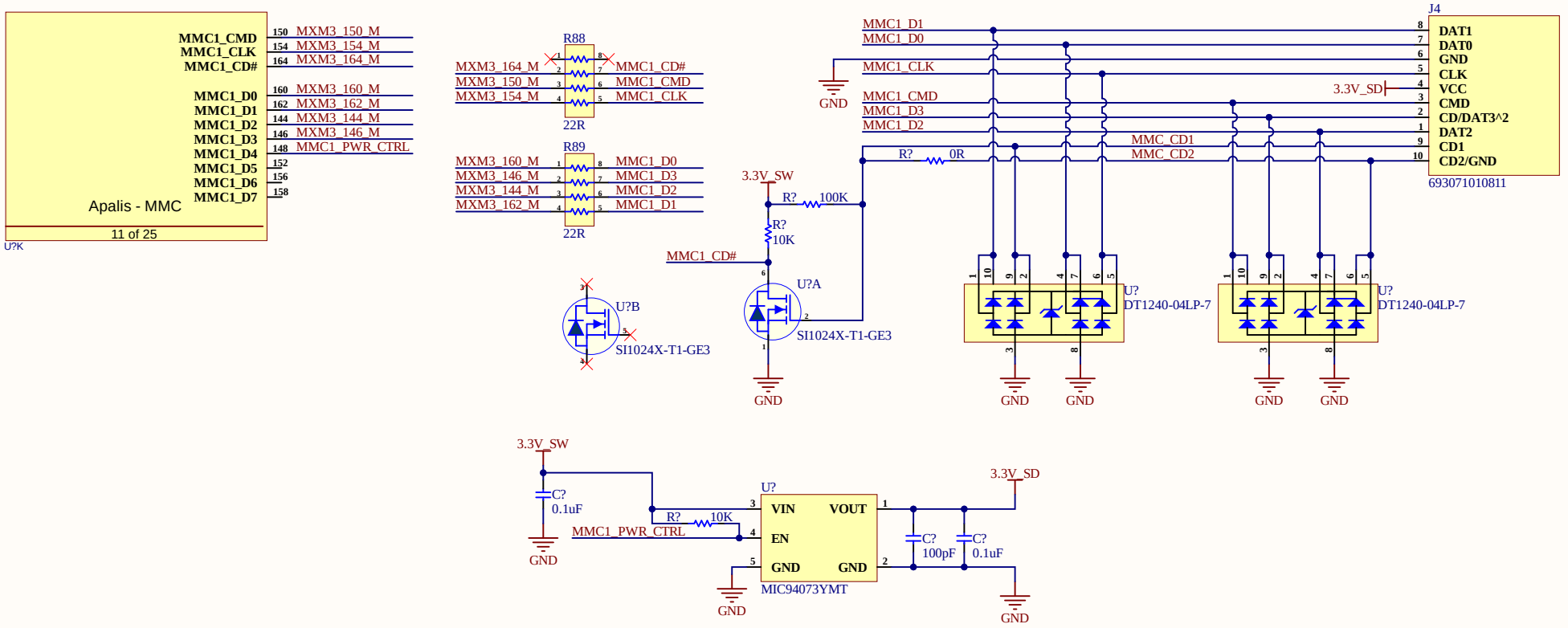

U3
MCP251863T-H/SSVAO
VDD
VIO
VCC
CANH
TXD
RXD
RXCAN
CANL
TXCAN
STBY
nCS
SCK
OSC2
SDI
OSC1
SDO
nINT
NC
nINT1/GPIO1
NC
nINT0/GPIO0/XSTBY
NC
CLKO/SOF
NC
NC
NC
GND
VSS

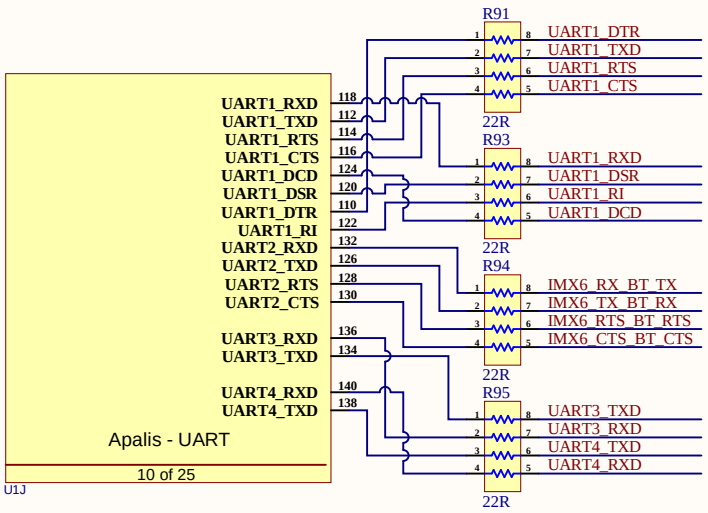

U?
VCC1
GND1
TXD
NC
RXD
NC
NC
GND1
VCC2
GND2
NC
CANH
CANL
VCC2
GND2
GND2
ISO1042QDWRQ1
CAN1_RX
CAN1_TX
CAN2_RX
CAN2_TX
Apalis - CAN
17 of 25
U?Q
MXM3_12_M
MXM3_14_M
MXM3_16_M
MXM3_18_M
R116
R117
R118
R119
22R
CAN1_RX_ISO
CAN1_TX_ISO
CAN1_RX
CAN1_TX
3.3V
5V
R155
0R
R156
DNP
C125
0.1uF
GND
U5
VCC
GND
TXD
RXD
RS
CANH
CANL
EN
SN65HVD234QDRQ1
R8
DNP
R153
DNP
CAN1_EN
R9
4.7K
L6
PE-1812ACC510STS
R12
DNP
R10
60.4R
R11
60.4R
C25
0.1uF
CAN1_HI
CAN1_LO
D?
DESD2CAN2SOQ-7


ETH1_MDI0+
ETH1_MDI0-
ETH1_MDI1+
ETH1_MDI1-
ETH1_MDI2+
ETH1_MDI2-
ETH1_MDI3+
ETH1_MDI3-
ETH1_ACT
ETH1_LINK
ETH1_CTREF
Apalis - Gigabit Ethernet
2 of 25
U1B
50 ETH1_MDI0_P
48 ETH1_MDI0_N
56 ETH1_MDI1_P
54 ETH1_MDI1_N
32 ETH1_MDI2_P
34 ETH1_MDI2_N
38 ETH1_MDI3_P
40 ETH1_MDI3_N
42 ETH1_ACT
44 ETH1_LINK
46 ETH1_CTREF
J1
TRCT3
TRD3-
TRD3+
TRD2+
TRD2-
TRCT2
TRCT4
TRD4+
TRD4-
TRD1-
TRD1+
TRCT1
K_YELLOW
A_YELLOW
K_ORANGE
COMMON
K_GREEN
SHIELD
SHIELD
A829-1J1T-KM
ETH1_CTREF_2
ETH1_MDI2_N
ETH1_MDI2_P
ETH1_MDI1_P
ETH1_MDI1_N
ETH1_CTREF_1
ETH1_CTREF_3
ETH1_MDI3_P
ETH1_MDI3_N
ETH1_MDI0_N
ETH1_MDI0_P
ETH1_CTREF_0
R14 150R
ETH1_ACT
C33 0.1uF
R15 DNP
3.3V_SW
R16 150R
ETH1_LINK
C34 1nF
R17 1M
GND
DNP COMP
R18 0R
C35 0.1uF
ETH1_CTREF_1
R19 0R
C36 0.1uF
ETH1_CTREF_2
ETH1_CTREF
L7
C37 47uF
C38 0.1uF
R20 0R
C39 0.1uF
ETH1_CTREF_3
R21 0R
C40 0.1uF
ETH1_CTREF_0

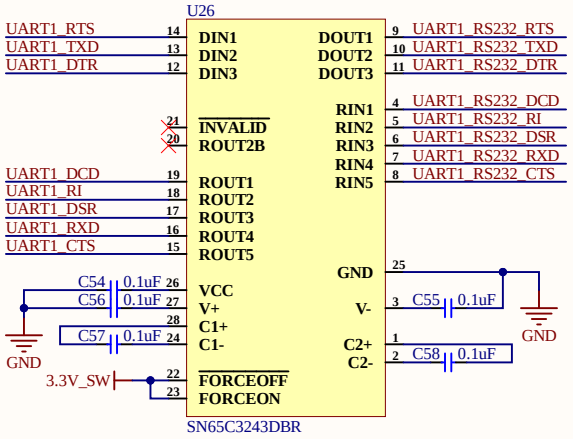

HDMI1_CEC
HDMI1_HPD
HDMI1_TXC-
HDMI1_TXC+
HDMI1_TXD0-
HDMI1_TXD0+
HDMI1_TXD1-
HDMI1_TXD1+
HDMI1_TXD2-
HDMI1_TXD2+
Apalis - HDMI
24 of 25
U1X
220 HDMI1_CEC_A
232 HDMI1_HPD_A
242 HDMI1_TXC_CON_N
240 HDMI1_TXC_CON_P
236 HDMI1_TXD0_CON_N
234 HDMI1_TXD0_CON_P
230 HDMI1_TXD1_CON_N
228 HDMI1_TXD1_CON_P
224 HDMI1_TXD2_CON_N
222 HDMI1_TXD2_CON_P
I2C2_SDA
I2C2_SCL
Apalis - Display DDC
25 of 25
U1Y
205
207
R90
22R
R23
4.7K
R24
4.7K
3.3V_SW
5V_SW
R142
0R
C43
0.1uF
GND
HDMI_CT_HPD
HDMI_LS_OE
3.3V_SW
C41
0.1uF
GND
U23
TPD12S016PWR
LS_OE
CT_HPD
VCCA
HPD_A-
HPD_B
CLK-
CLK+
D0-
D0+
D1-
D1+
D2-
D2+
CEC_A
SCL_A
SDA_A
CEC_B
SCL_B
SDA_B
VCC5V
5V_OUT
GND
HDMI1_HPD_A
HDMI1_CEC_A
HDMI1_HPD_CON
HDMI1_TXC_CON_N
HDMI1_TXC_CON_P
HDMI1_TXD0_CON_N
HDMI1_TXD0_CON_P
HDMI1_TXD1_CON_N
HDMI1_TXD1_CON_P
HDMI1_TXD2_CON_N
HDMI1_TXD2_CON_P
HDMI1_CEC_CON
HDMI1_DDC_SCL
HDMI1_DDC_SDA
HDMI1_VDISP
C44
0.1uF
GND
J2
MX50B19NQ1
MH1
MH2
MH3
MH4
MH5
MH6
MH7
MH8
C42
1nF
R22
1M
GND
CH_GND
HDMI1_VDISP
GND
5V_SW
5V_SW

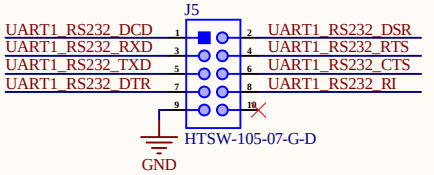

Apalis - USB
3 of 25
U1C
USBO1_D-
USBO1_D+
USBO1_SSTX-
USBO1_SSTX+
USBO1_SSRX-
USBO1_SSRX+
USBO1_VBUS
USBO1_ID
USBO1_OC#
USBO1_EN
USBH_OC#
USBH_EN
USBH2_D-
USBH2_D+
USBH3_D-
USBH3_D+
USBH4_D-
USBH4_D+
USBH4_SSTX-
USBH4_SSTX+
USBH4_SSRX-
USBH4_SSRX+
USBO1_D_N
USBO1_D_P
USBO1_SSTX_N
USBO1_SSTX_P
USBO1_SSRX_N
USBO1_SSRX_P
R133 22R_USB01_OC#
R134 22R_USBO1_EN
R132 22R_USBH_OC#
R131 22R_USBH_EN
LTE_USB_D-_N
LTE_USB_D+_P
USBH4_D_N
USBH4_D_P
USBH4_SSTX_N
USBH4_SSTX_P
USBH4_SSRX_N
USBH4_SSRX_P
U7
TPD4E02B04QDQARQ1
U8
TPD4E02B04QDQARQ1
U9
TPD4E02B04QDQARQ1
U10
TPD4E02B04QDQARQ1
NC NC NC NC
GND GND
IO1 IO2 IO3 IO4
GND
J3
STDA_SSRX-_U1
STDA_SSRX+_U1
GND_DRAIN_U1
STDA_SSTX-_U1
STDA_SSTX+_U1
VBUS_U1
D-_U1
D+_U1
GND_U1
STDA_SSRX-_U2
STDA_SSRX+_U2
GND_DRAIN_U2
STDA_SSTX-_U2
STDA_SSTX+_U2
VBUS_U2
D-_U2
D+_U2
GND_U2
SHIELD
0484060003
VCC_USBH4
VCC_USBO1
L8
L9
L10
L11
MCZ1210DH900L2TA0G
AGND
GND
C45 1nF
R25 1M
5V_SW
R144 0R
U11A
U11B
IN OUT1
EN1 OC1
EN2 OUT2
OC2 GND
TPS2066TDGNRQ1
USBH_EN
USBH_OC#
USBO1_EN
USB01_OC#
L12
L13
C46 0.1uF
C47 100uF
C48 1nF
C49 0.1uF
C50 100uF
C51 1nF


MMC1_CMD
MMC1_CLK
MMC1_CD#
MMC1_D0
MMC1_D1
MMC1_D2
MMC1_D3
MMC1_D4
MMC1_D5
MMC1_D6
MMC1_D7
Apalis - MMC
11 of 25
U?K
150 MXM3_150_M
154 MXM3_154_M
164 MXM3_164_M
160 MXM3_160_M
162 MXM3_162_M
144 MXM3_144_M
146 MXM3_146_M
148 MMC1_PWR_CTRL
152
156
158
R88
MXM3_164_M
MXM3_150_M
MXM3_154_M
MMC1_CD#
MMC1_CMD
MMC1_CLK
22R
R89
MXM3_160_M
MXM3_146_M
MXM3_144_M
MXM3_162_M
MMC1_D0
MMC1_D3
MMC1_D2
MMC1_D1
22R
GND
MMC1_D1
MMC1_D0
MMC1_CLK
MMC1_CMD
MMC1_D3
MMC1_D2
R? 0R
MMC_CD1
MMC_CD2
3.3V_SD
J4
DAT1
DAT0
GND
CLK
VCC
CMD
CD/DAT3^2
DAT2
CD1
CD2/GND
693071010811
3.3V_SW
R? 100K
R? 10K
MMC1_CD#
U?A
SI1024X-T1-GE3
U?B
SI1024X-T1-GE3
U?
DT1240-04LP-7
U?
DT1240-04LP-7
3.3V_SW
C? 0.1uF
R? 10K
MMC1_PWR_CTRL
U?
VIN
VOUT
EN
GND
GND
MIC94073YMT
C? 100pF
C? 0.1uF
3.3V_SD

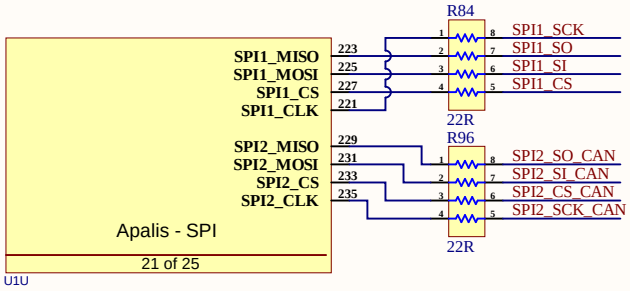


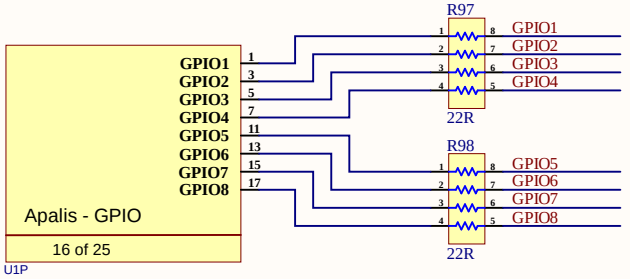
UART1 - FOR DEBUG
UART2 - FOR BT
UART3 - FOR FUTURE USE
UART4 - FOR FUTURE USE

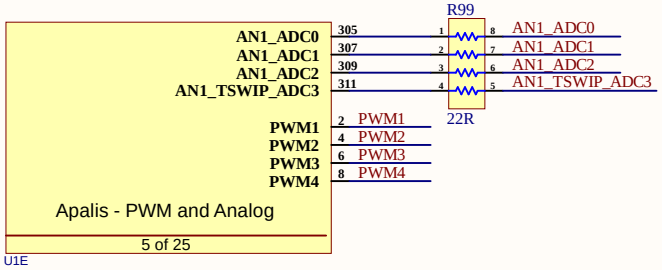




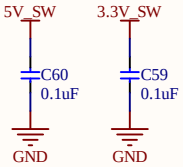
CONNECTOR PENDING

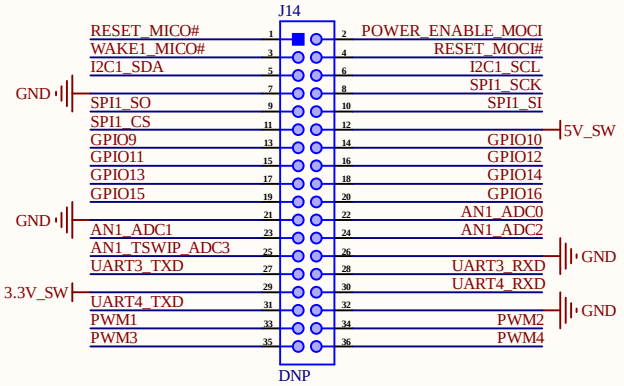

R84
SPI1_MISO
SPI1_MOSI
SPI1_CS
SPI1_CLK
223
225
227
221
SPI1_SCK
SPI1_SO
SPI1_SI
SPI1_CS
22R
R96
SPI2_MISO
SPI2_MOSI
SPI2_CS
SPI2_CLK
229
231
233
235
SPI2_SO_CAN
SPI2_SI_CAN
SPI2_CS_CAN
SPI2_SCK_CAN
22R
Apalis - SPI
21 of 25
U1U

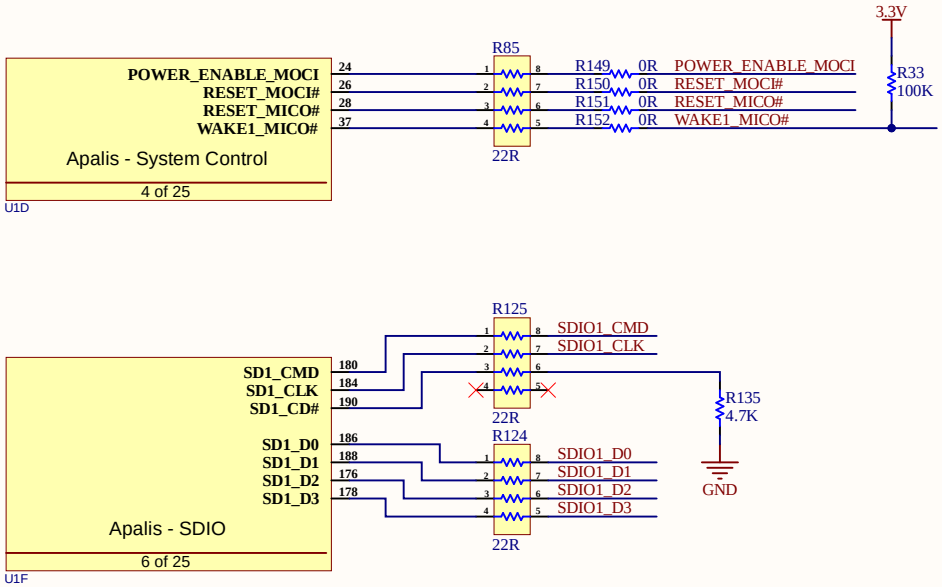

R97
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
1
3
5
7
11
13
15
17
GPIO1
GPIO2
GPIO3
GPIO4
22R
R98
GPIO5
GPIO6
GPIO7
GPIO8
22R
Apalis - GPIO
16 of 25
U1P

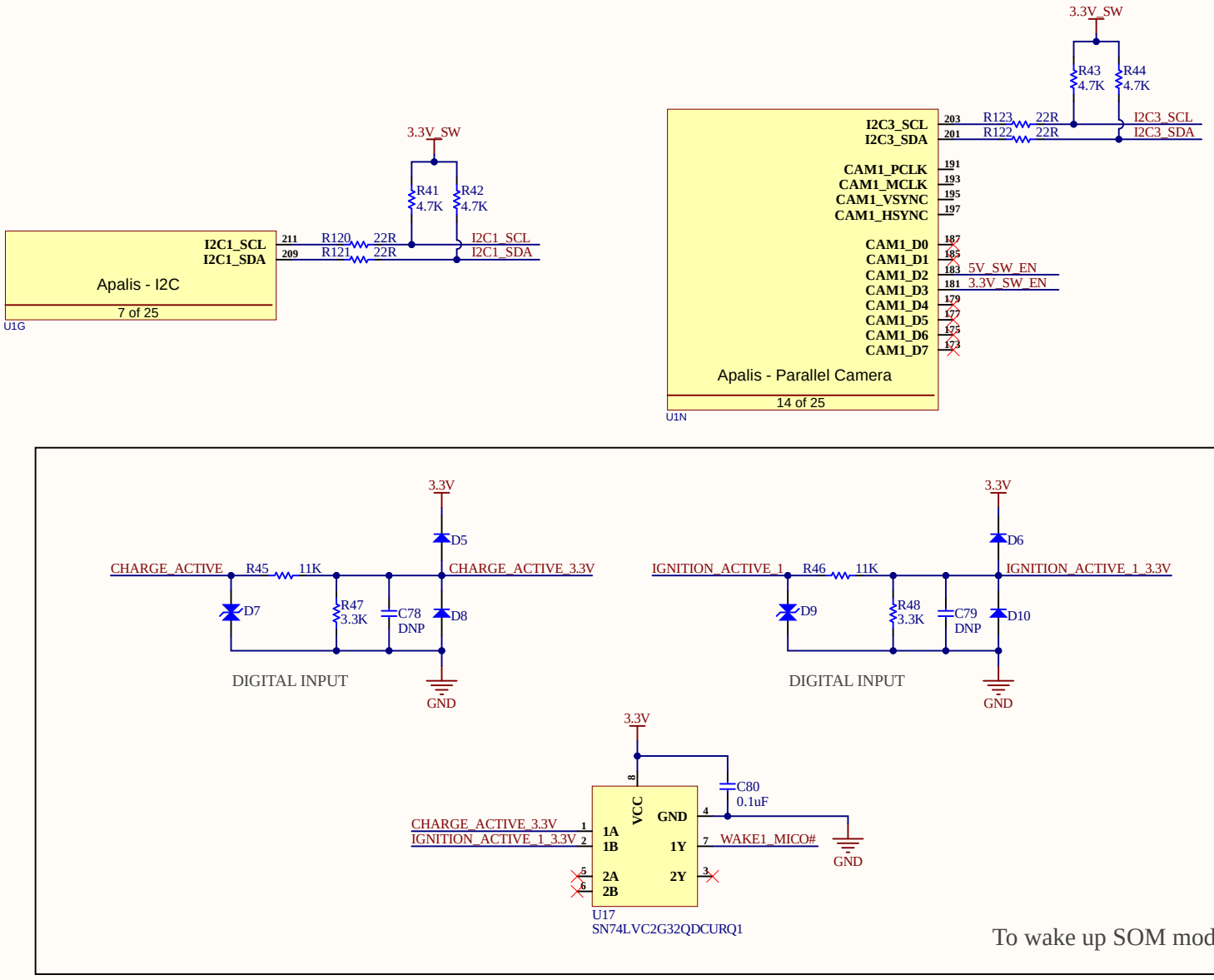

R99
AN1_ADC0
AN1_ADC1
AN1_ADC2
AN1_TSWIP_ADC3
305
307
309
311
AN1_ADC0
AN1_ADC1
AN1_ADC2
AN1_TSWIP_ADC3
22R
PWM1
PWM2
PWM3
PWM4
2 PWM1
4 PWM2
6 PWM3
8 PWM4
Apalis - PWM and Analog
5 of 25
U1E

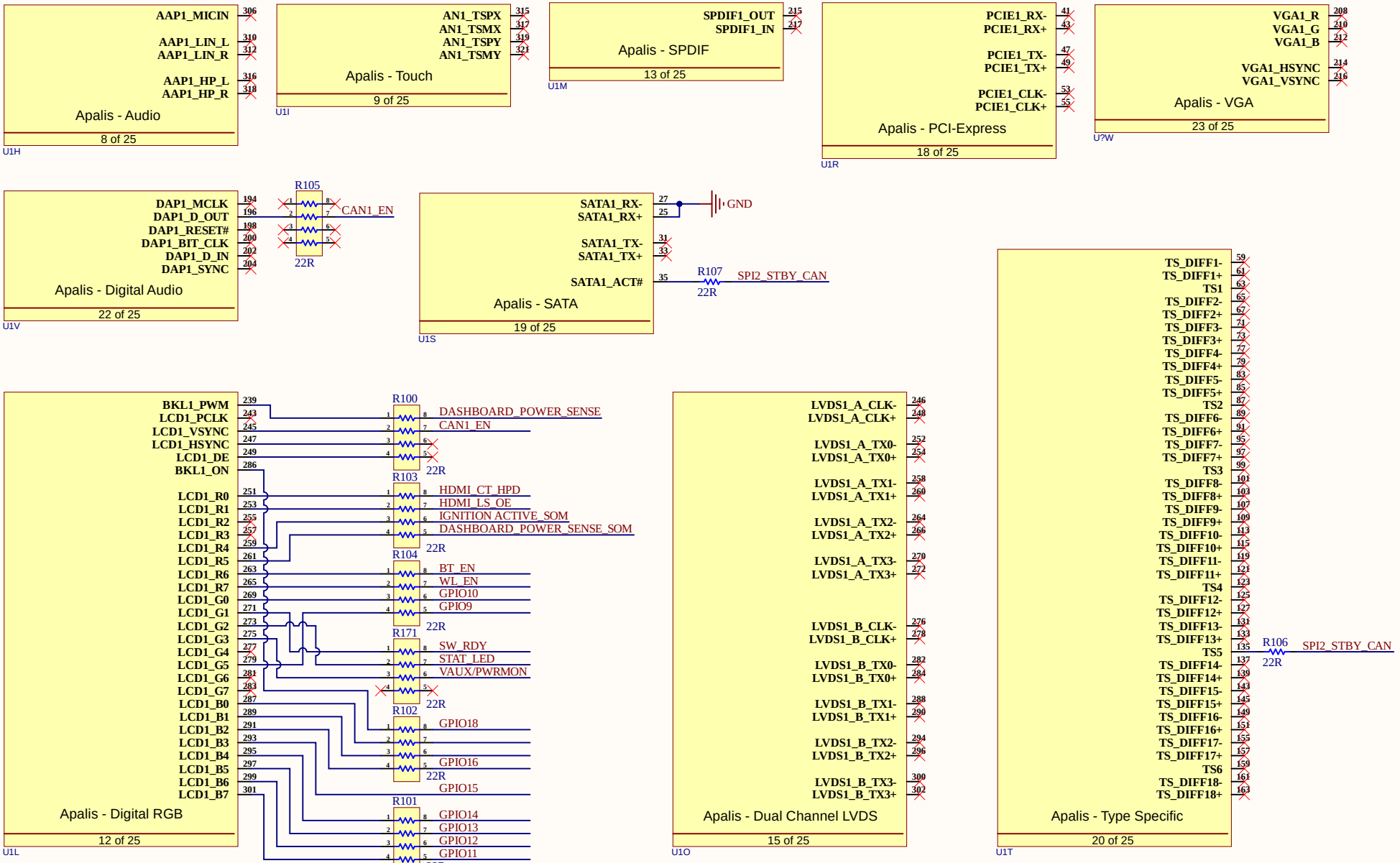

3.3V_SW
5V_SW
3.3V_SW
5V_SW

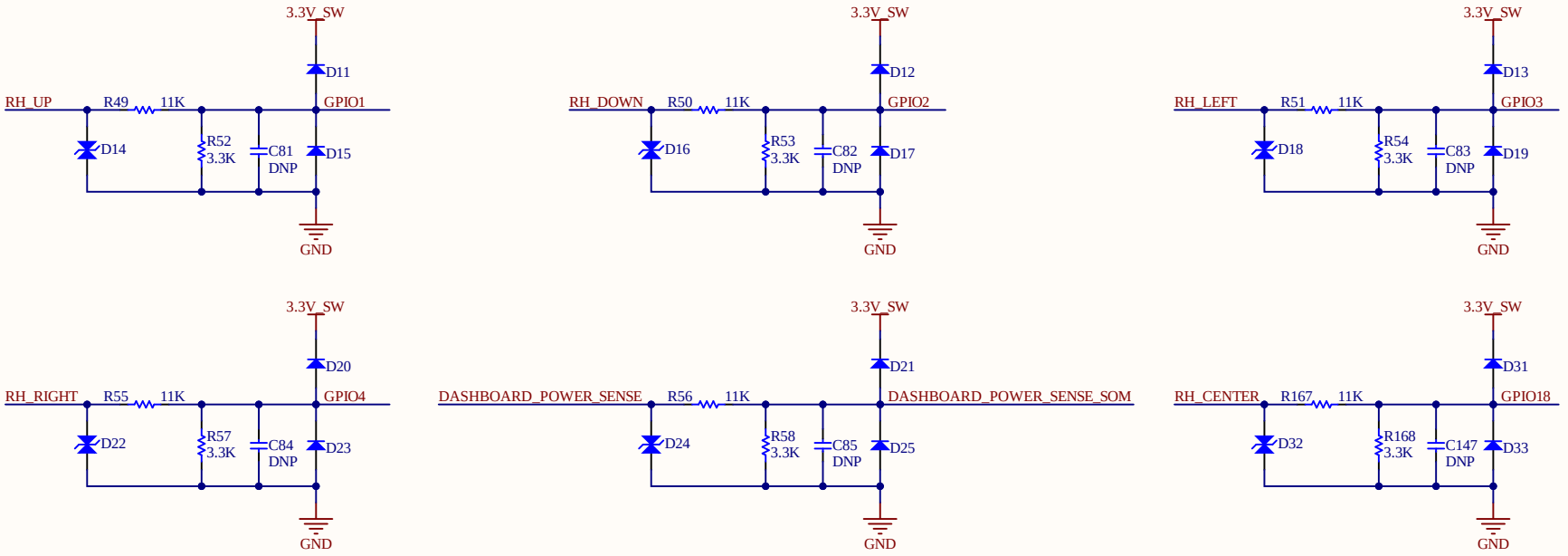

5V_SW
C60
0.1uF
GND
3.3V_SW
C59
0.1uF
GND


J14
RESET_MICO#
WAKE1_MICO#
I2C1_SDA
GND
SPI1_SO
SPI1_CS
GPIO9
GPIO11
GPIO13
GPIO15
GND
AN1_ADC1
AN1_TSWIP_ADC3
UART3_TXD
3.3V_SW
UART4_TXD
PWM1
PWM3
POWER_ENABLE_MOCI
RESET_MOCI#
I2C1_SCL
SPI1_SCK
SPI1_SI
5V_SW
GPIO10
GPIO12
GPIO14
GPIO16
AN1_ADC0
AN1_ADC2
GND
UART3_RXD
UART4_RXD
GND
PWM2
PWM4
DNP

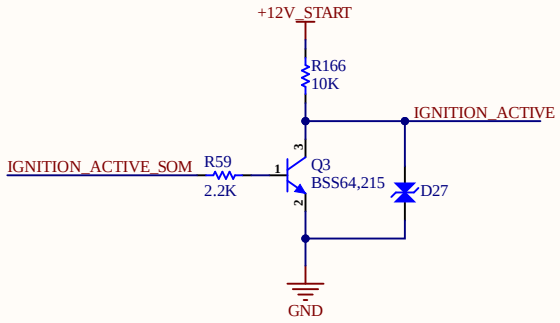

POWER_ENABLE_MOCI
RESET_MOCI#
RESET_MICO#
WAKE1_MICO#
Apalis - System Control
4 of 25
U1D
24
26
28
37
R85
22R
R149 0R POWER_ENABLE_MOCI
R150 0R RESET_MOCI#
R151 0R RESET_MICO#
R152 0R WAKE1_MICO#
3.3V
R33
100K
SD1_CMD
SD1_CLK
SD1_CD#
SD1_D0
SD1_D1
SD1_D2
SD1_D3
Apalis - SDIO
6 of 25
U1F
180
184
190
186
188
176
178
R125
R124
22R
22R
SDIO1_CMD
SDIO1_CLK
SDIO1_D0
SDIO1_D1
SDIO1_D2
SDIO1_D3
R135
4.7K
GND


3.3V_SW
R43
4.7K
R44
4.7K
I2C3_SCL
I2C3_SDA
203
201
R123
22R
R122
22R
I2C3_SCL
I2C3_SDA
CAM1_PCLK
CAM1_MCLK
CAM1_VSYNC
CAM1_HSYNC
CAM1_D0
CAM1_D1
CAM1_D2
CAM1_D3
CAM1_D4
CAM1_D5
CAM1_D6
CAM1_D7
191
193
195
197
187
185
183
181
179
177
175
173
5V_SW_EN
3.3V_SW_EN
Apalis - Parallel Camera
14 of 25
U1N
3.3V_SW
R41
4.7K
R42
4.7K
I2C1_SCL
I2C1_SDA
211
209
R120
22R
R121
22R
I2C1_SCL
I2C1_SDA
Apalis - I2C
7 of 25
U1G
3.3V
D5
CHARGE_ACTIVE
R45
11K
CHARGE_ACTIVE_3.3V
D7
R47
3.3K
C78
DNP
D8
DIGITAL INPUT
GND
3.3V
D6
IGNITION_ACTIVE_1
R46
11K
IGNITION_ACTIVE_1_3.3V
D9
R48
3.3K
C79
DNP
D10
DIGITAL INPUT
GND
3.3V
8
C80
0.1uF
CHARGE_ACTIVE_3.3V
IGNITION_ACTIVE_1_3.3V
1
2
1A
1B
VCC
GND
4
1Y
7
WAKE1_MICO#
2A
2B
5
6
2Y
3
GND
U17
SN74LVC2G32QDCURQ1
To wake up SOM module

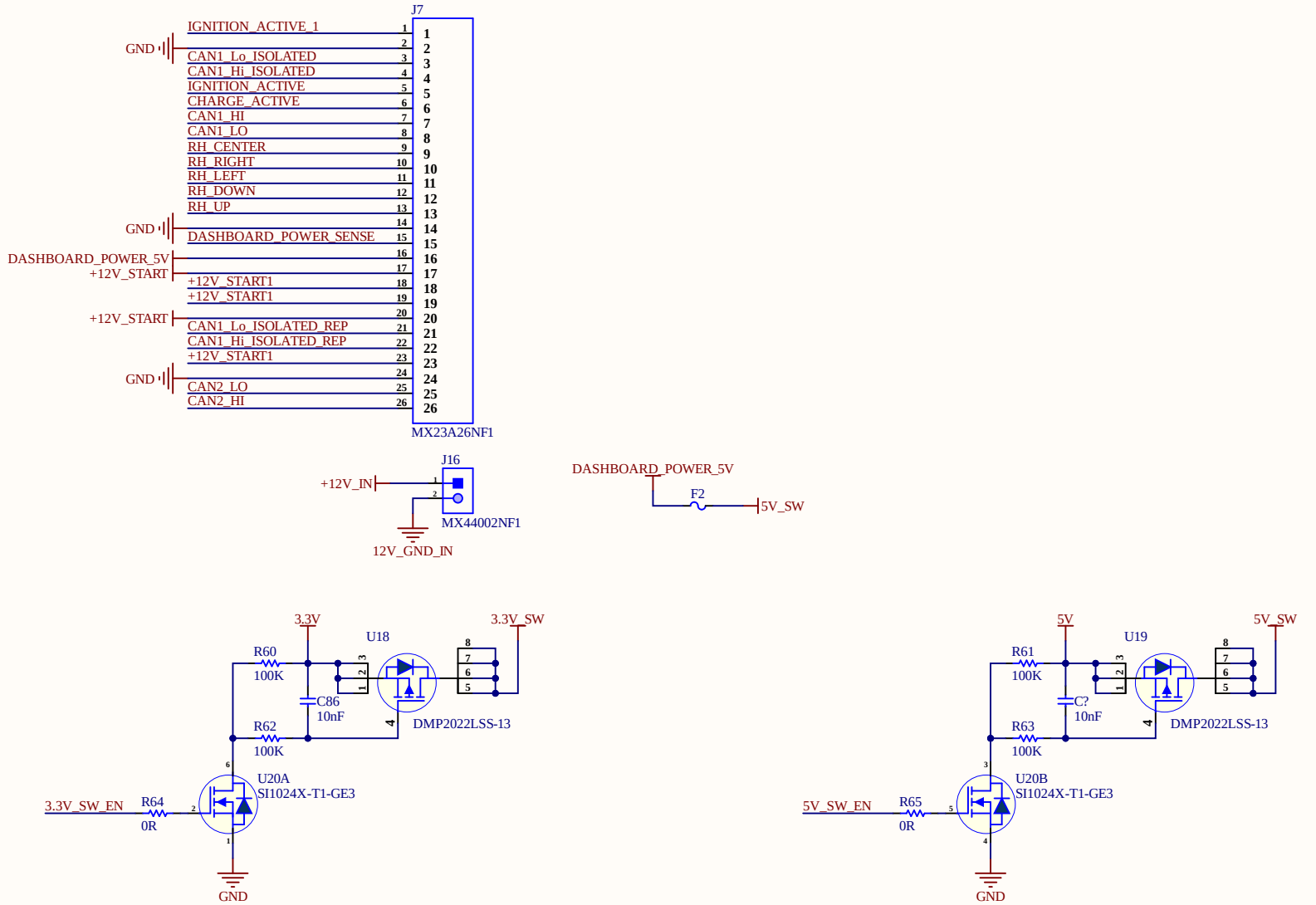

AAP1_MICIN
AAP1_LIN_L
AAP1_LIN_R
AAP1_HP_L
AAP1_HP_R
Apalis - Audio
8 of 25
U1H
AN1_TSPX
AN1_TSMX
AN1_TSPY
AN1_TSMY
Apalis - Touch
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U1I
SPDIF1_OUT
SPDIF1_IN
Apalis - SPDIF
13 of 25
U1M
PCIE1_RX-
PCIE1_RX+
PCIE1_TX-
PCIE1_TX+
PCIE1_CLK-
PCIE1_CLK+
Apalis - PCI-Express
18 of 25
U1R
VGA1_R
VGA1_G
VGA1_B
VGA1_HSYNC
VGA1_VSYNC
Apalis - VGA
23 of 25
U?W
DAP1_MCLK
DAP1_D_OUT
DAP1_RESET#
DAP1_BIT_CLK
DAP1_D_IN
DAP1_SYNC
Apalis - Digital Audio
22 of 25
U1V
R105
22R
CAN1_EN
SATA1_RX-
SATA1_RX+
SATA1_TX-
SATA1_TX+
SATA1_ACT#
GND
R107
22R
SPI2_STBY_CAN
Apalis - SATA
19 of 25
U1S
TS_DIFF1-
TS_DIFF1+
TS1
TS_DIFF2-
TS_DIFF2+
TS_DIFF3-
TS_DIFF3+
TS_DIFF4-
TS_DIFF4+
TS_DIFF5-
TS_DIFF5+
TS2
TS_DIFF6-
TS_DIFF6+
TS_DIFF7-
TS_DIFF7+
TS3
TS_DIFF8-
TS_DIFF8+
TS_DIFF9-
TS_DIFF9+
TS_DIFF10-
TS_DIFF10+
TS_DIFF11-
TS_DIFF11+
TS4
TS_DIFF12-
TS_DIFF12+
TS_DIFF13-
TS_DIFF13+
TS5
TS_DIFF14-
TS_DIFF14+
TS_DIFF15-
TS_DIFF15+
TS_DIFF16-
TS_DIFF16+
TS_DIFF17-
TS_DIFF17+
TS6
TS_DIFF18-
TS_DIFF18+
R106
22R
SPI2_STBY_CAN
Apalis - Type Specific
20 of 25
U1T
BKL1_PWM
LCD1_PCLK
LCD1_VSYNC
LCD1_HSYNC
LCD1_DE
BKL1_ON
LCD1_R0
LCD1_R1
LCD1_R2
LCD1_R3
LCD1_R4
LCD1_R5
LCD1_R6
LCD1_R7
LCD1_G0
LCD1_G1
LCD1_G2
LCD1_G3
LCD1_G4
LCD1_G5
LCD1_G6
LCD1_G7
LCD1_B0
LCD1_B1
LCD1_B2
LCD1_B3
LCD1_B4
LCD1_B5
LCD1_B6
LCD1_B7
Apalis - Digital RGB
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U1L
R100
DASHBOARD_POWER_SENSE
CAN1_EN
22R
R103
HDMI_CT_HPD
HDMI_LS_OE
IGNITION ACTIVE_SOM
DASHBOARD_POWER_SENSE_SOM
22R
R104
BT_EN
WL_EN
GPIO10
GPIO9
22R
R171
SW_RDY
STAT_LED
VAUX/PWRMON
22R
R102
GPIO18
GPIO16
22R
GPIO15
R101
GPIO14
GPIO13
GPIO12
GPIO11
22R
LVDS1_A_CLK-
LVDS1_A_CLK+
LVDS1_A_TX0-
LVDS1_A_TX0+
LVDS1_A_TX1-
LVDS1_A_TX1+
LVDS1_A_TX2-
LVDS1_A_TX2+
LVDS1_A_TX3-
LVDS1_A_TX3+
LVDS1_B_CLK-
LVDS1_B_CLK+
LVDS1_B_TX0-
LVDS1_B_TX0+
LVDS1_B_TX1-
LVDS1_B_TX1+
LVDS1_B_TX2-
LVDS1_B_TX2+
LVDS1_B_TX3-
LVDS1_B_TX3+
Apalis - Dual Channel LVDS
15 of 25
U1O

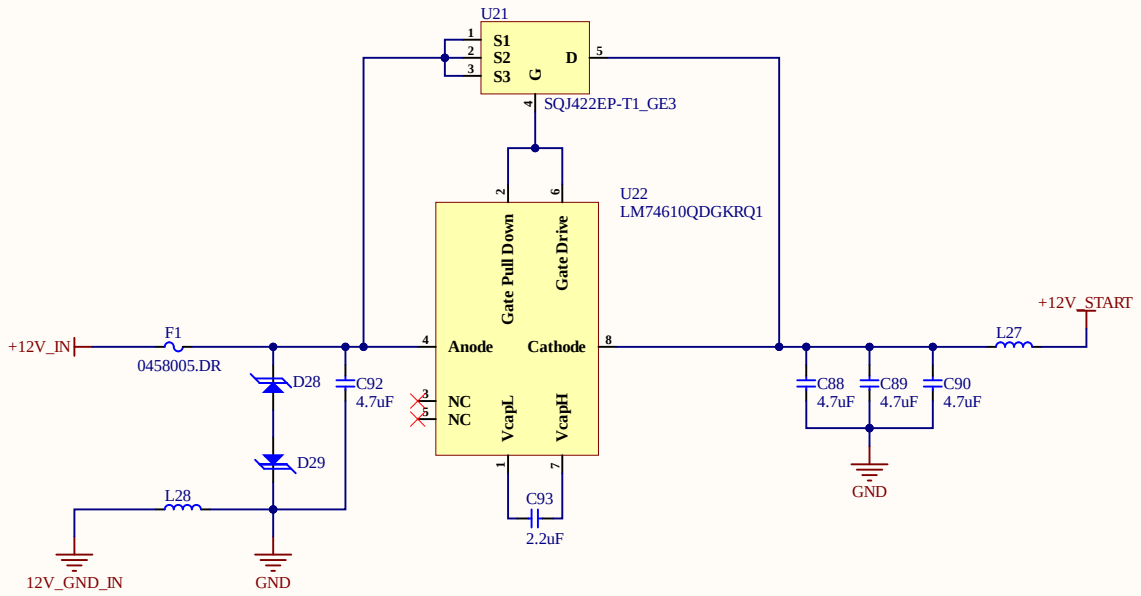


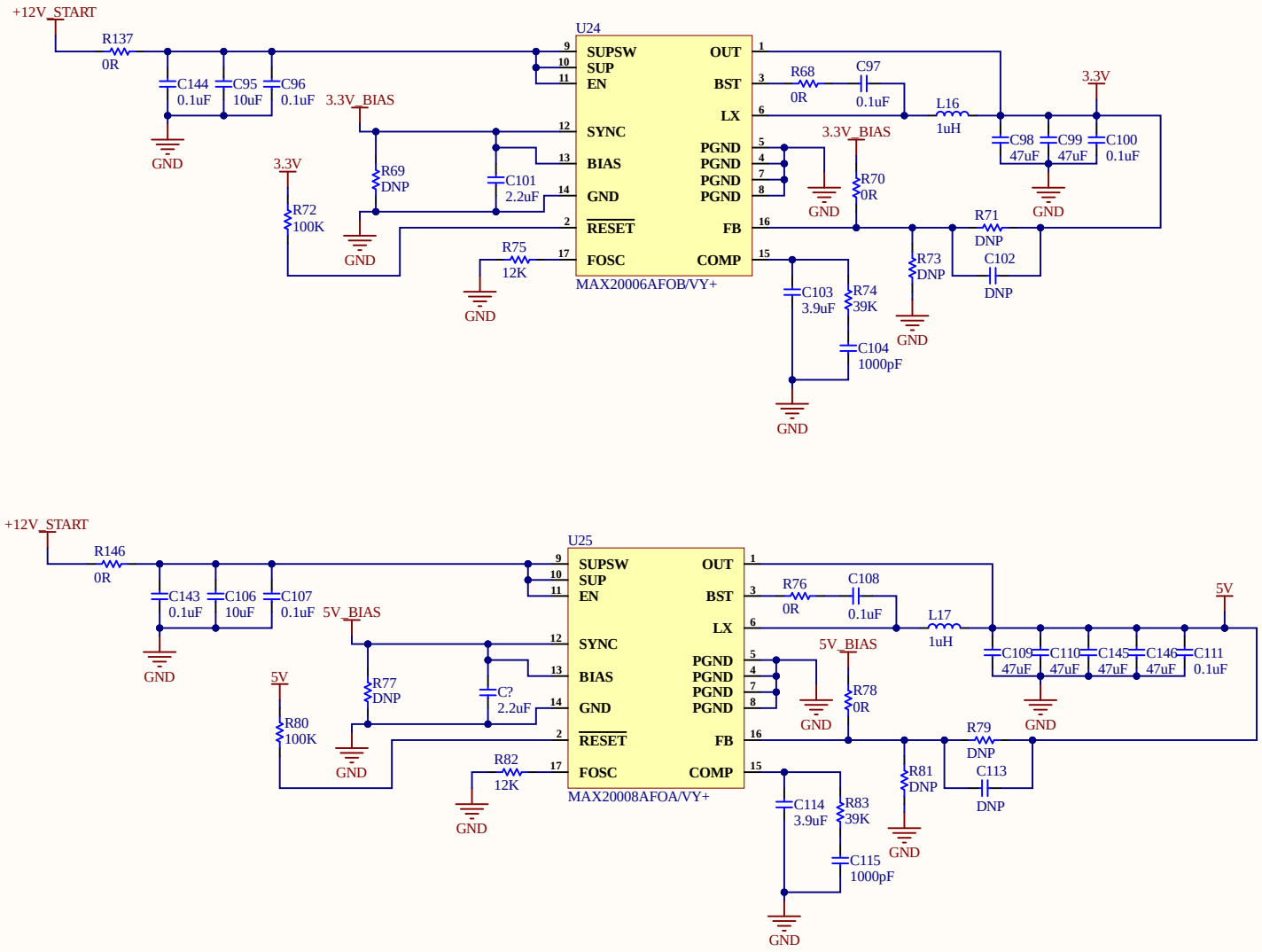
DIGITAL INPUT

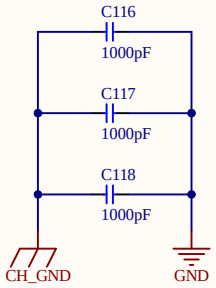


DIGITAL OUTPUT


J7
IGNITION_ACTIVE_1
GND
CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED
IGNITION_ACTIVE
CHARGE_ACTIVE
CAN1_HI
CAN1_LO
RH_CENTER
RH_RIGHT
RH_LEFT
RH_DOWN
RH_UP
GND
DASHBOARD_POWER_SENSE
DASHBOARD_POWER_5V
+12V_START
+12V_START1
+12V_START1
+12V_START
CAN1_Lo_ISOLATED_REP
CAN1_Hi_ISOLATED_REP
+12V_START1
GND
CAN2_LO
CAN2_HI
MX23A26NF1
J16
+12V_IN
MX44002NF1
12V_GND_IN
DASHBOARD_POWER_5V
F2
5V_SW
3.3V
U18
3.3V_SW
R60
100K
C86
10nF
R62
100K
DMP2022LSS-13
U20A
SI1024X-T1-GE3
3.3V_SW_EN
R64
0R
GND
5V
U19
5V_SW
R61
100K
C?
10nF
R63
100K
DMP2022LSS-13
U20B
SI1024X-T1-GE3
5V_SW_EN
R65
0R
GND


U21
S1
S2
S3
G
D
SQJ422EP-T1_GE3
U22
LM74610QDGKRQ1
Gate Pull Down
Gate Drive
Anode
Cathode
NC
NC
VcapL
VcapH
+12V_IN
F1
0458005.DR
D28
D29
C92
4.7uF
L28
12V_GND_IN
GND
C93
2.2uF
C88
4.7uF
C89
4.7uF
C90
4.7uF
L27
+12V_START
GND


+12V_START
R137
0R
C144
0.1uF
C95
10uF
C96
0.1uF
GND
3.3V_BIAS
3.3V
R69
DNP
R72
100K
C101
2.2uF
R75
12K
U24
SUPSW
SUP
EN
SYNC
BIAS
GND
RESET
FOSC
OUT
BST
LX
PGND
PGND
PGND
PGND
FB
COMP
MAX20006AFOB/VY+
R68
0R
C97
0.1uF
L16
1uH
3.3V
C98
47uF
C99
47uF
C100
0.1uF
R70
0R
R71
DNP
C102
DNP
R73
DNP
C103
3.9uF
R74
39K
C104
1000pF
+12V_START
R146
0R
C143
0.1uF
C106
10uF
C107
0.1uF
5V_BIAS
5V
R77
DNP
R80
100K
C?
2.2uF
R82
12K
U25
MAX20008AFOA/VY+
R76
0R
C108
0.1uF
L17
1uH
5V
C109
47uF
C110
47uF
C145
47uF
C146
47uF
C111
0.1uF
R78
0R
R79
DNP
C113
DNP
R81
DNP
C114
3.9uF
R83
39K
C115
1000pF


C116
1000pF
C117
1000pF
C118
1000pF
CH_GND
GND